CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

4.2 Equilibrium of Forces

Syllabus Section: 4. Forces, Density & Pressure **Paper Type:** Structured Questions (Paper 2) **Total:** 10 Questions (106 Marks)

Question 1 Easy [10 marks]

Part (a) [3 marks]

Define the following

(i) moment of a force

[2]

(ii) centre of gravity.

[1]

Mark Scheme (3 marks):

(i) A moment of a force is defined as:

- The turning effect of a force; *[1 mark]*
- Moment = Force × perpendicular distance from the pivot **OR** $M = Fd$; *[1 mark]*

(ii) Centre of gravity is defined as:

- The single point where the weight (of a body) can be taken as acting from; *[1 mark]*

[Total: 3 marks]

Part (b) [3 marks]

A teacher sets some Physics students the task of finding the weight of a broom using only a ruler, some thin, light string, and a set of laboratory masses. The equipment is initially arranged as shown in Fig. 1.1.

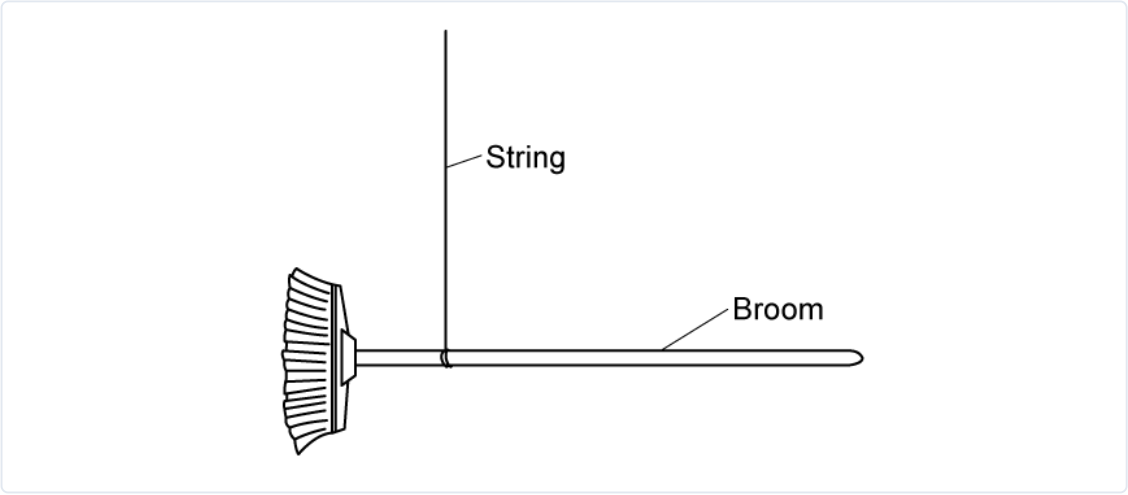


Fig. 1.1.

The students are not allowed to use scales.

For the arrangement in Fig. 1.1.

(i) State why the broom is hanging horizontally.

[2]

(ii) State what measurement the students should make to start their investigation to find the weight of the broom.

[1]

Mark Scheme (3 marks):

For the arrangement in Fig. 1.1.

(i) State why the broom is hanging horizontally:

- The broom is in equilibrium; *[1 mark]*
- The string is fixed at the centre of mass; *[1 mark]*

(ii) State what measurement the students should make to start their investigation to find the weight of the broom:

- Distance from either end to the string; *[1 mark]*

[Total: 3 marks]

The students will need to know the position of the centre of mass, so they definitely need the distance from one end to the string, which must be at the centre of mass for equilibrium.

Part (c) **[2 marks]**

The measurement to the position of the string found in part (b) is shown in Fig. 1.2.

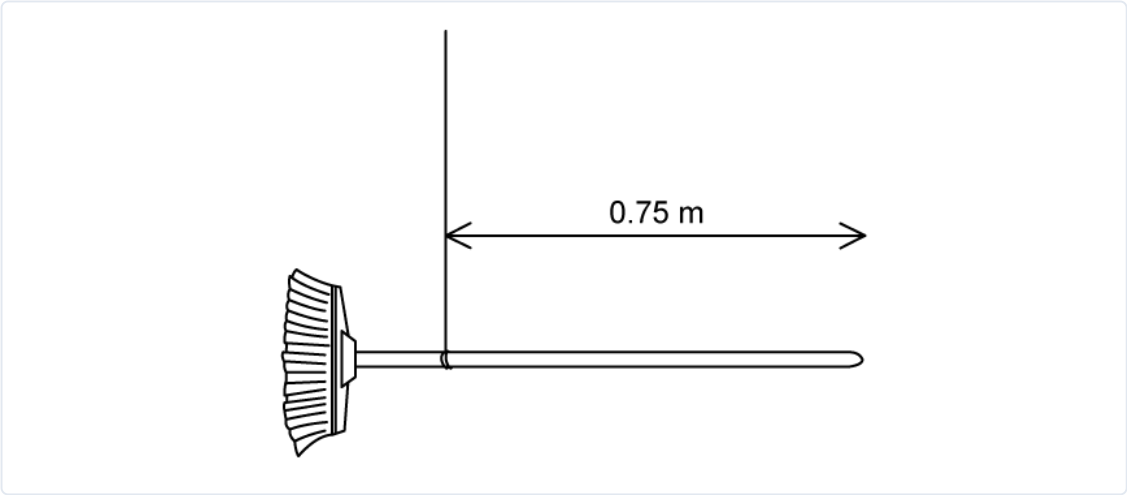


Fig. 1.2.

Distance from string to the end of the handle = 0.75 m

The string is moved towards the end of the handle, so that the broom is in equilibrium when a 0.5 kg mass is hung a distance of 0.45 m from the string as shown in Fig. 1.3.

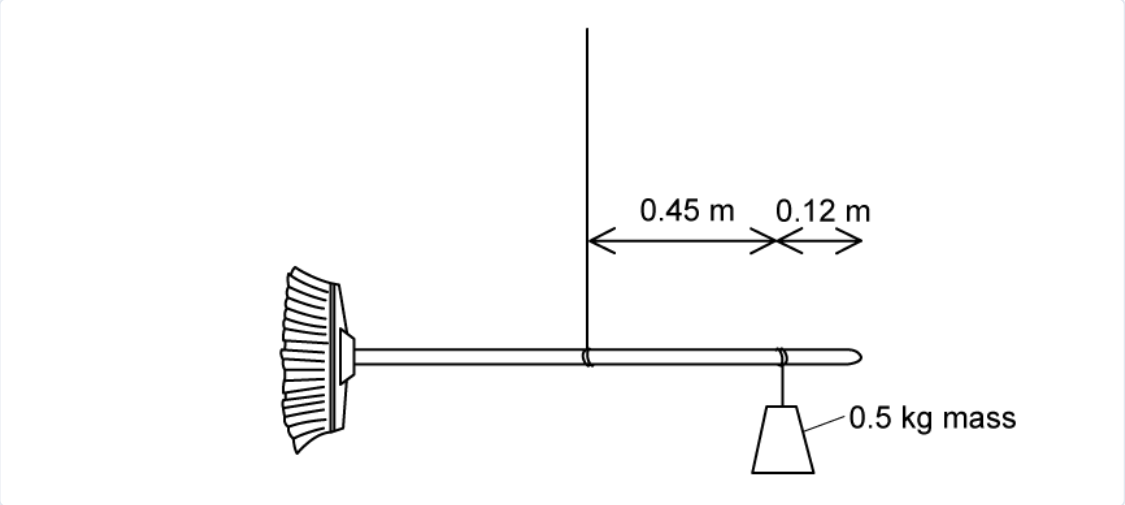


Fig. 1.3

Determine

(i) the distance between the centre of mass of the broom and the point where the string is attached.

[1]

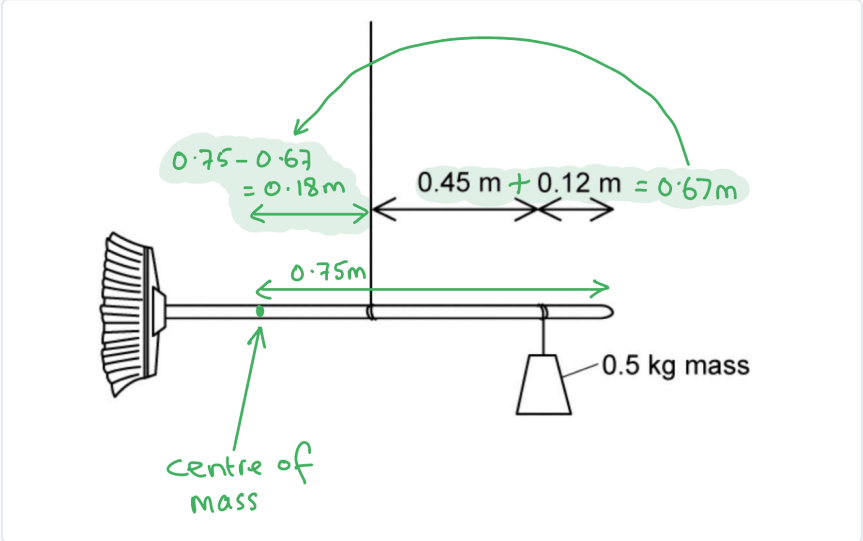
(ii) the force exerted by the 0.5 kg mass.

[1]

Mark Scheme (2 marks):

(i) Determine the distance between the centre of mass of the broom and the point where the string is attached:

- Total distance from end of handle to centre of mass = 0.75 m



- Distance to centre of mass, $d = 0.75 - 0.45 + 0.12 = 0.18$ m; *[1 mark]*

(ii) Determine the force exerted by the 0.5 kg mass:

- force exerted = weight acting downwards
- $W = mg = 0.5 \times 9.81 = 4.9$ N *[1 mark]*

[Total: 2 marks]

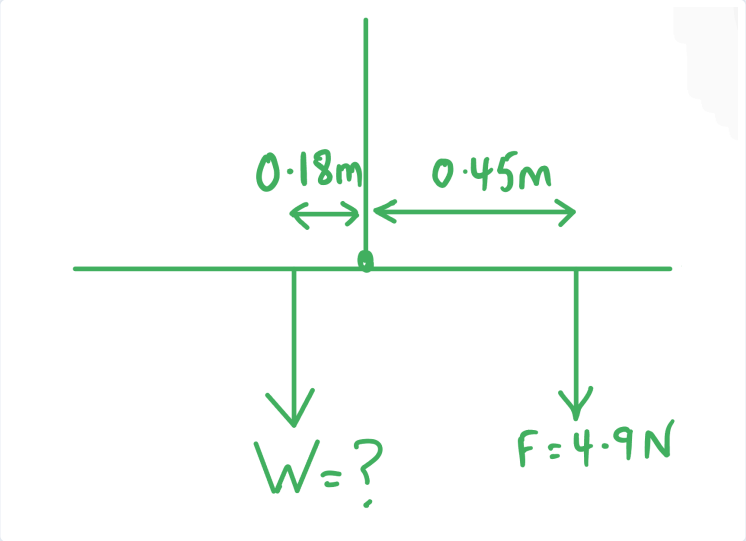
Part (d) [2 marks]

Calculate the weight of the broom.

Mark Scheme (2 marks):

Calculate the weight of the broom:

Sketch a diagram to show the forces and distances:



Apply the principle of moments to the point where the string is attached to the broom:

- Sum of anti-clockwise moments = sum of clockwise moments
- $W \times 0.18 = 4.9 \times 0.45$ *[1 mark]*
- $W = \frac{4.9 \times 0.45}{0.18} = 12.25$
- $W = 12$ N 2 s.f. *[1 mark]*

[Total: 2 marks]

Many students are reluctant to 'waste time' sketching diagrams which don't get marks. This is not a waste of time!

The diagram ensures you write the correct values in your calculation.

Although not marked directly, it is an essential part of the process to find the solution.

Question 2 Easy [9 marks]

Part (a) [3 marks]

A metre rule is balanced on a pivot at its mid-point.

A force of 10.0 N acts at a distance of 8.0 cm from one end of the rule as shown in Fig. 1.1.

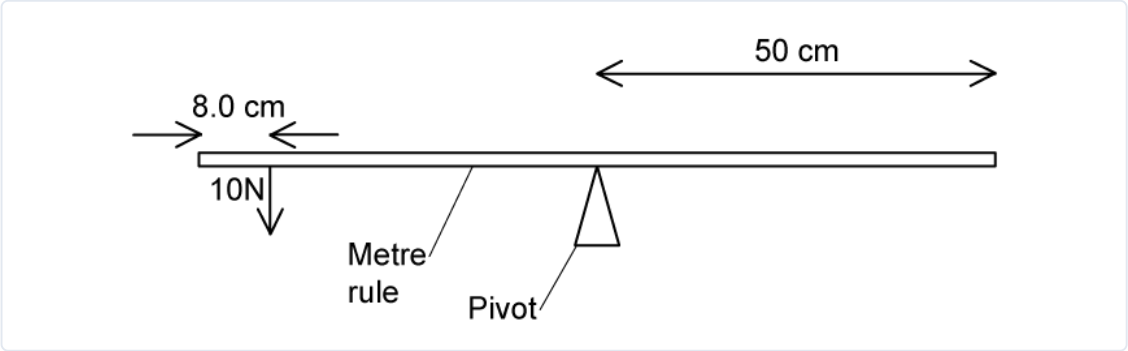


Fig. 1.1

Calculate the moment of the 10.0 N force about the pivot.

moment = ... N m

Mark Scheme (3 marks):

To calculate the moment:

List the known quantities:

- Force, F = 10 N
- Distance of force from the end of the ruler = 8.0 cm
- Therefore distance of force from pivot, $d = 50 - 8 = 42$ cm or 0.42 m *[1 mark]*

Recall the equation for a moment and calculate:

- $M = Fd$
- $M = 10 \times 0.42$ *[1 mark]*
- $M = 4.2$ N m *[1 mark]*

[Total: 3 marks]

In a question like this one the first mark is for identifying the correct distance, so at that stage you could answer in cm or m and still get the mark.

*However the question specifies N**m** as the final unit, so to get the next two marks you have to convert to **metres**.*

Keep an eye on units in moments questions as they can be tricky.

Part (b) [2 marks]

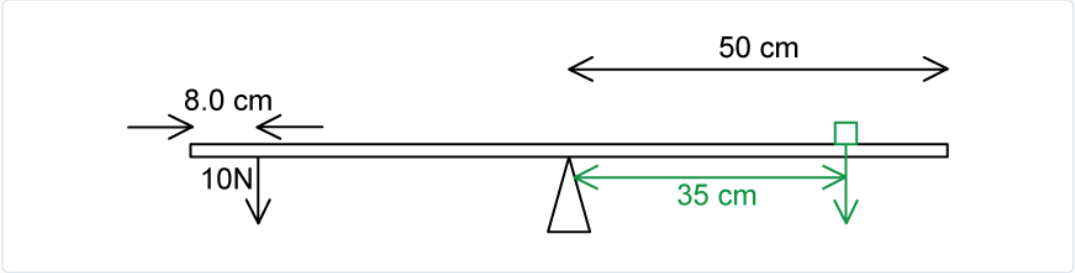
A mass is placed at a point 35 cm from the pivot, which makes the metre rule balance.

On Fig. 1.1, draw an arrow to show the position of the mass and the direction of the force it exerts.

Mark Scheme (2 marks):

On Fig. 1.1, draw an arrow to show the position of the mass and the direction of the force it exerts.

A correctly annotated diagram looks like:



- Downwards arrow marked on the right of the pivot (so it causes a clockwise moment); *[1 mark]*
- Mass / force arrow 35 cm from pivot (distance labelled or by eye); *[1 mark]*

[Total: 2 marks]

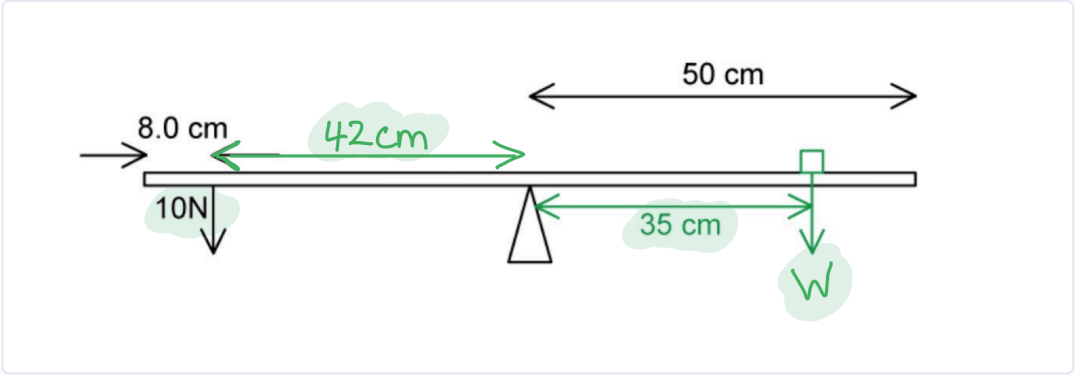
Part (c) **[4 marks]**

Calculate the **mass** which has been placed to balance the ruler in part (a).

Mark Scheme (4 marks):

Calculate the **mass** which has been placed to balance the ruler in part (a):

Use the diagram and distances from parts (a) and (b):



Apply the principle of moments since the ruler is in equilibrium:

- sum of clockwise moments = sum of anti-clockwise moments
- $W \times 35 = 10 \times 42$ *[1 mark]*
- $W = \frac{420}{35} = 12$ N *[1 mark]*

Convert weight to mass:

- $m = \frac{W}{g} = \frac{12}{9.81}$ *[1 mark]*
- $m = 1.2 \text{ kg}$ *[1 mark]*

[Total: 4 marks]

Question 3 Medium [10 marks]

Part (a) [6 marks]

A team of decorators set up a raised bench to allow them to reach the ceiling.

The bench consists of a plank which is resting in a horizontal position on two supports **B** and **C**. The plank is modelled as a uniform rod **AD** of length 2.7 m and mass 15 kg. The supports at **B** and **C** are 0.4 m from each end of the plank, as shown in Fig 1.1.

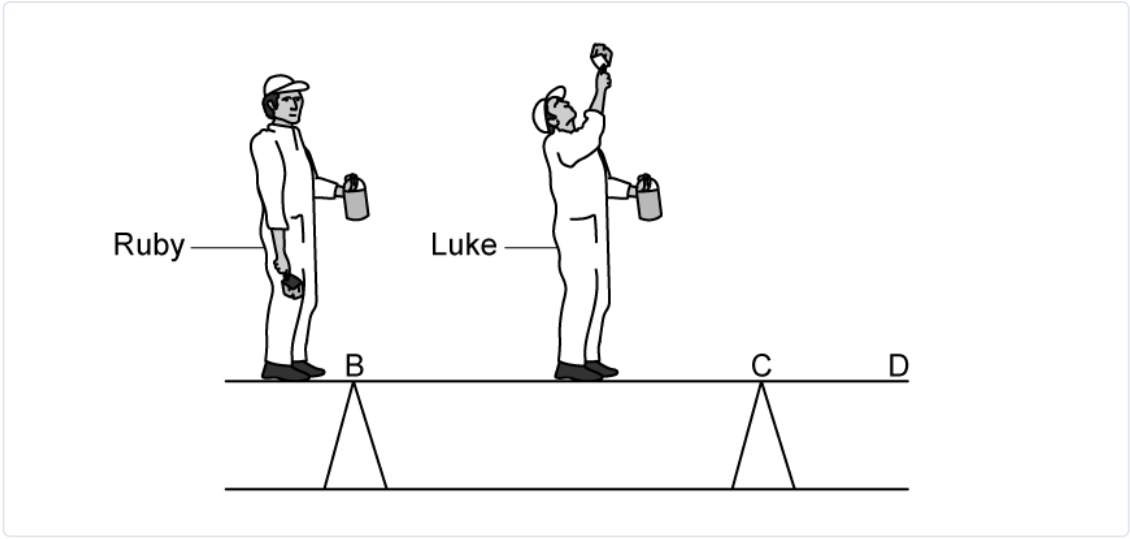


Fig 1.1

Two decorators, Ruby and Luke, stand on the plank. Luke has a mass of 85 kg and stands in the middle of the plank.

Ruby has a mass of 72 kg and stands at end **A**.

The plank remains horizontal and in equilibrium. The decorators can be modelled as point masses.

Calculate:

(i) The magnitude of the normal reaction force at **B**.

[3]

(ii) The magnitude of the normal reaction force at **C**.

[3]

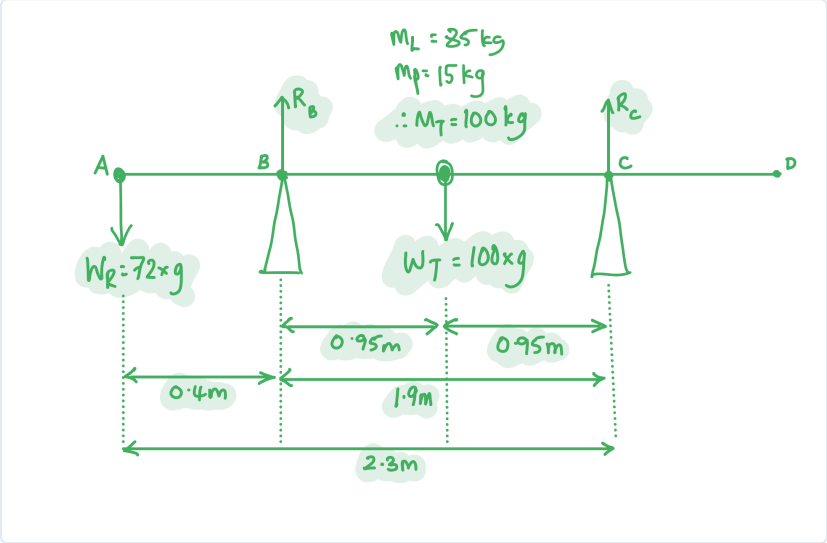
Mark Scheme (6 marks):

List the known quantities:

- Mass of Ruby, M_R = 72 kg
- Mass of Luke, M_L = 85 kg

- Mass of the plank, M_P = 15 kg
- Length of plank, AD = 2.7 m
- Acceleration due to gravity, g = 9.81 m s^{-2}

Sketch a simplified diagram showing measurements and point masses:



(i) Calculate the magnitude of the normal reaction force at **B**

Apply the principle of moments to take moments about point **C**:

- sum of clockwise moments = sum of anticlockwise moments
- $R_B \times 1.9 = W_R \times 2.7 - 0.4 + W_P + W_L \times 0.95$ *[1 mark]*
- $R_B = \frac{72g \times 2.3 + 100g \times 0.95}{1.9}$ *[1 mark]*
- $R_B = 1346 = 1300$ N 2 s.f. *[1 mark]*

(ii) Calculate the magnitude of the normal reaction force at **C**

Equate vertical forces:

- Since the plank is in equilibrium, the sum of upwards forces upwards = sum of forces downwards / $\Sigma F_{up} = \Sigma F_{down}$ / $R_B + R_C = W_R + W_L + W_P$ *[1 mark]*

Calculate normal reaction force at **C**:

- $R_C = W_R + W_L + W_P - R_B$
- $R_C = 72g + 85g + 15g - 1346$ *[1 mark]*
- $R_C = 341 = 340$ N 2 s.f. *[1 mark]*

[Total: 6 marks]

It's very common for students to think that creating their own diagrams is a waste of time since 'it doesn't get any marks'.

*It's true that the diagram isn't marked by the examiner directly, **but** it allows you to get the correct answer quickly, meaning you not only get these marks, but have **time** to answer the rest of the paper too!*

For part (ii) you may have decided to take moments again. Any correct method that gets a correct answer will score full marks in the exam.

If you chose this method, your calculation should look like:

- *Apply the principle of moments to take moments about point **B**:*
  - *sum of clockwise moments = sum of anticlockwise moments*
  - $W_P + W_L \times 0.95 = R_C \times 1.9 + W_R \times 0.4$ *[1 mark]*
  - $R_C = \frac{100g \times 0.95 - 72g \times 0.4}{1.9}$ *[1 mark]*
  - $R_C = 342 = 340 \text{ N } 2 \text{ s.f.}$ *[1 mark]*

Part (b) **[4 marks]**

Whilst Ruby stays at point **A**, Luke now moves along the plank to a point **X**.

The plank remains horizontal and in equilibrium, and the magnitude of the normal reaction force at **B** is now twice the magnitude of the normal reaction force at **C**.

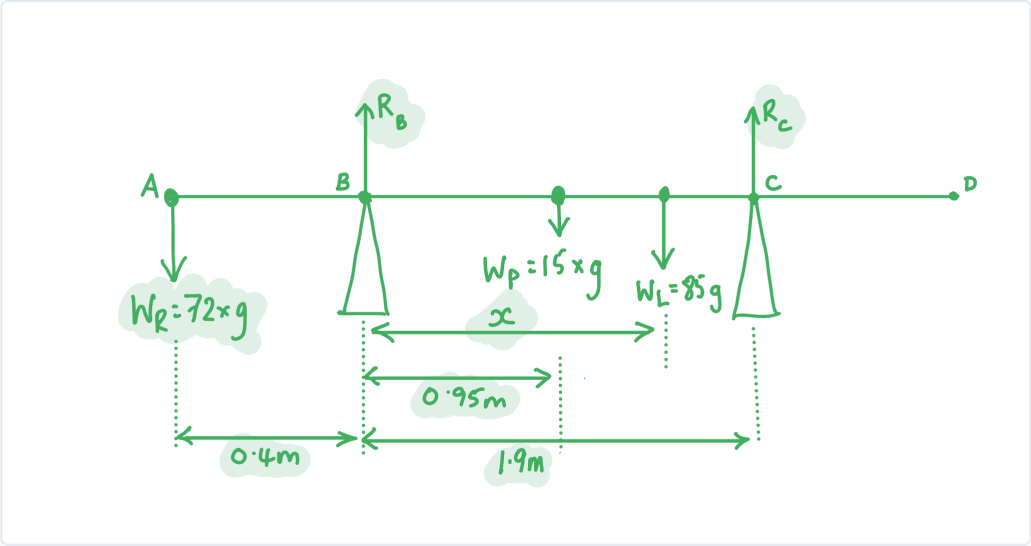
Calculate the distance **BX**.

Mark Scheme (4 marks):

List the known quantities:

- Mass of Ruby, M_R = 72 kg
- Mass of Luke, M_L = 85 kg
- Mass of the plank, M_P = 15 kg
- Length of plank, **AD** = 2.7 m
- Acceleration due to gravity, g = 9.81 m s^{-2}

Sketch a simplified diagram showing the new position of Luke:



Equate vertical forces:

- Since the plank is in equilibrium, the sum of upwards forces upwards = sum of forces downwards / $\Sigma F_{up} = \Sigma F_{down}$ / $R_B + R_C = W_R + W_L + W_P$

Write an expression, in terms of R_C, the reaction force at **C**:

- $2 \times R_C + R_C = W_R + W_L + W_P$
- $3 \times R_C = W_R + W_L + W_P$ *[1 mark]*
- $R_C = \frac{72g + 85g + 15g}{3} = 562.44$
- $R_C = 560$ N 2 s.f. *[1 mark]*

Determine distance **BX** by taking moments about **B**:

- $W_P \times 0.95 + W_L \times x = R_C \times 1.9 + W_R \times 0.4$
- $x = \frac{R_C \times 1.9 + W_R \times 0.4 - W_P \times 0.95}{W_L}$
- $x = \frac{560 \times 1.9 + 72 \times 9.81 \times 0.4 - 15 \times 9.81 \times 0.95}{85 \times 9.81}$ *[1 mark]*
- $x = 1.45 = 1.5$ m 2 s.f. *[1 mark]*

[Total: 4 marks]

Question 4 Medium [14 marks]

Part (a) [2 marks]

A student investigates the principle of moments using the apparatus shown in Fig. 1.1.

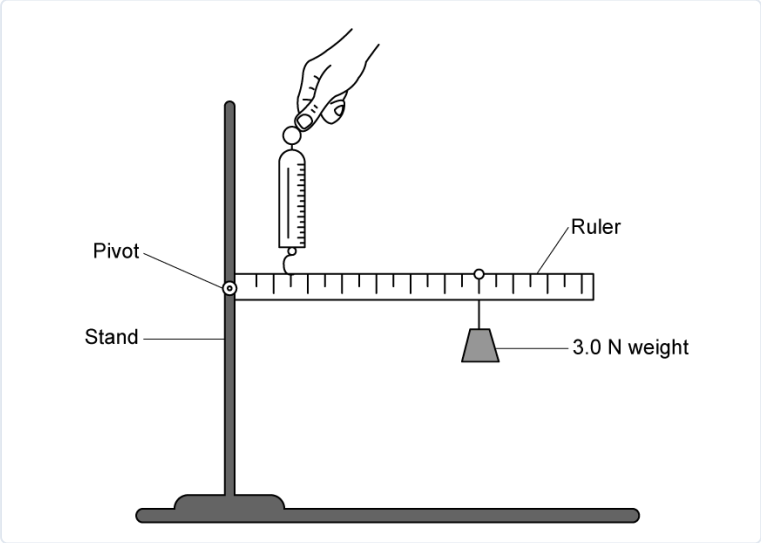


Fig. 1.1

The scale on the forcemeter reads from 0 N to 10 N.

A 3.0 N weight is positioned so that it hangs straight downwards from the 70 cm mark on the ruler.

Describe what the student could do to ensure that the ruler is horizontal.

Mark Scheme (2 marks):

Describe what the student could do to ensure that the ruler is horizontal:

*Any one **pair** from:*

- Measure height from bench to ruler in several places; *[1 mark]*
- Check the measurements are all the same; *[1 mark]*
- Place spirit level on the ruler; *[1 mark]*
- Bubble should be central; *[1 mark]*
- Measure angle between the ruler and the stand; *[1 mark]*
- Check it is a right angle; *[1 mark]*
- Place set square (triangle) at the join of the ruler and stand; *[1 mark]*
- Make sure the angle is a right angle; *[1 mark]*
- Set a marker level with the pivot; *[1 mark]*
- Ensure it is the same height as the end of the ruler; *[1 mark]*

[Total: 2 marks]

Part (b) **[4 marks]**

The student intends to investigate moments around the pivot.

(i) State the equation linking moment, force and distance from the pivot.

[2]

(ii) Calculate the moment of the 3.0 N weight.

[2]

Mark Scheme (4 marks):

The student intends to investigate moments around the pivot:

(i) State the equation linking moment, force and distance from the pivot:

- $M = F \times d$ **OR** moment = force × distance; *[1 mark]*
- Where F = force / applied load and d = perpendicular distance from the pivot; *[1 mark]*

(ii) Calculate the moment of the 3.0 N weight:

List the known quantities:

- F = 3.0 N
- d = 70 cm = 0.7 m

Calculate the moment and write final answer with units:

- $M = 3.0 \times 0.7$ *[1 mark]*
- $M = 2.1 \text{ Nm}$ *[1 mark]*

[Total: 4 marks]

*When calculating moments it is acceptable to calculate distance using centimetres rather than metres (unless stated otherwise), but you **must** state the unit, which would be N cm in that case.*

Part (c) **[5 marks]**

The student positions the forcemeter at the 15 cm mark and uses it to hold the ruler horizontal.

He expects the reading on the forcemeter to be 14 N.

The actual reading is 10 N.

(i) Infer, showing your calculation, why the student made this assumption.

[2]

(ii) Explain why the correct reading should be **larger** than 14 N.

[2]

(iii) Explain why the actual reading is only 10 N.

[1]

Mark Scheme (5 marks):

(i) Infer, showing your calculation, why the student made this assumption:

Apply the principle of moments:

- clockwise moment = anti-clockwise moment **OR** $2.1 \text{ Nm} = F_m \times 0.15 \text{ m}$ *[1 mark]*
- Force read from forcemeter, $F_m = \frac{2.1}{0.15} = 14 \text{ N}$ *[1 mark]*

(ii) Explain why the correct reading should be **larger** than 14 N:

- The mass / weight of the ruler; *[1 mark]*
- Acts in the clockwise direction / has been ignored; *[1 mark]*

(iii) Explain why the actual reading is only 10 N:

- The scale on the forcemeter only goes up to 10 N; *[1 mark]*

[Total: 5 marks]

*If you found yourself stuck for the final part of this question, then you had forgotten that in part (a) the question told you the forcemeter **only went up to 10 N**.*

*When you find yourself at a loss to answer any question it is **always** worth going back to the start and reading the previous subparts again. Often you will have forgotten small details which didn't seem important but are relevant now.*

Part (d) **[3 marks]**

The teacher asks the student and his friend to move a table with some books on it. They carry the table as shown in Fig. 1.2.

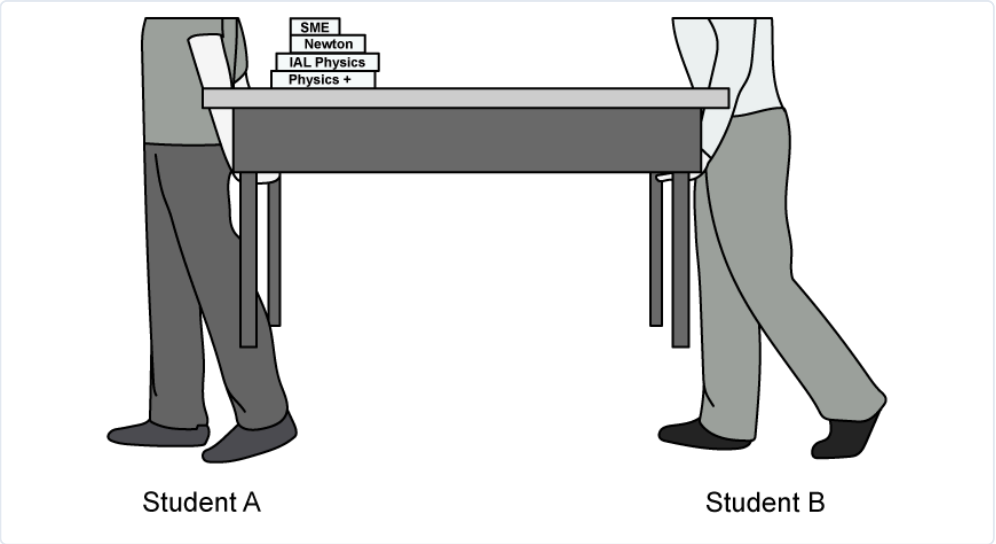


Fig. 1.2

Use ideas about moments to explain why student **A** feels more force on their hands than student **B**.

Mark Scheme (3 marks):

Student **A** feels more force on their hands than student **B** because:

- The desk is in equilibrium so the clockwise moment = anti-clockwise moment; *[1 mark]*
- The books are closer to **A** / further from **B OR** $d_A < d_B$; *[1 mark]*
- Therefore force at A is larger / force at B is smaller / $F_A > F_B$ / force and distance are inversely proportional to create the same moment; *[1 mark]*

[Total: 3 marks]

If you are writing your mark scheme with reference to $d_A < d_B$ and $F_A > F_B$, make sure these are correctly labelled on Fig 1.2 so the examiner can see exactly what you're referring to. You should avoid stating variables that you have no stated the meaning of or annotated on a diagram.

Question 5 Medium [8 marks]

Part (a) [5 marks]

A pole **PR** has a length 6.0 m and a weight *W* N. The pole is held in a horizontal position in equilibrium by two vertical ropes attached to the pole as shown in Fig. 1.1.

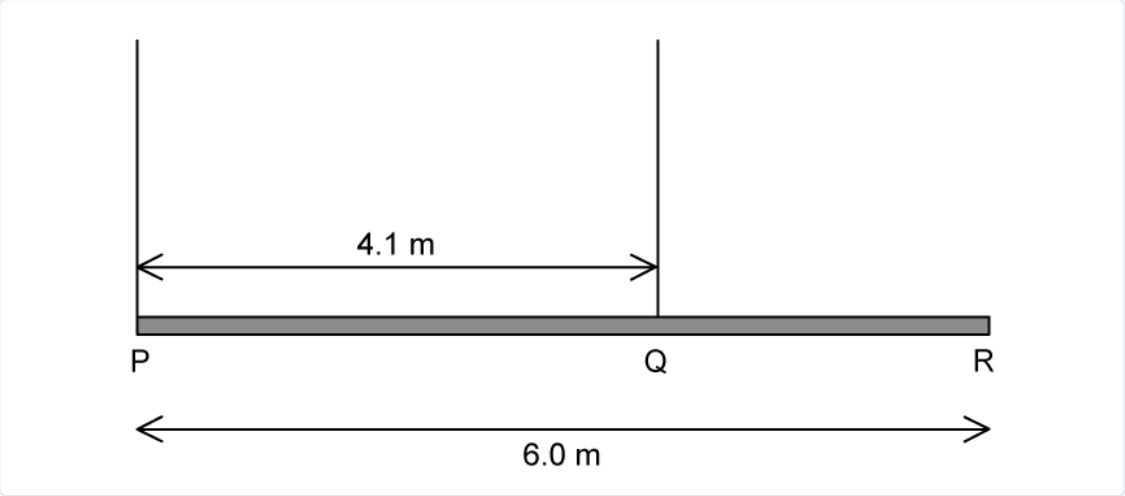


Fig. 1.1.

A load of weight 10 N is attached to the rod at **R**.

The tension in the rope attached at **Q** is six times as large as the tension in the rope attached at **P**.

The pole is uniform and the ropes are light inextensible strings.

Calculate the value of *W*.

Mark Scheme (5 marks):

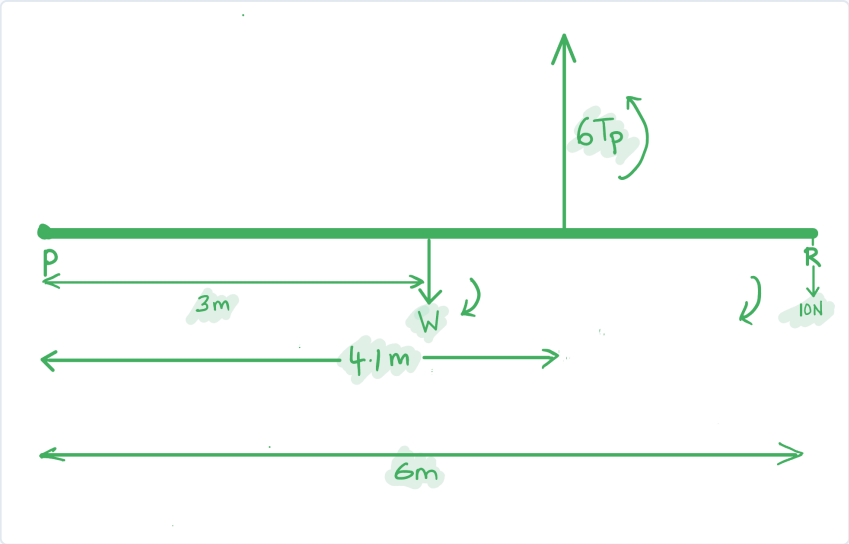
Calculate the value of *W*

Write an expression for T_P in terms of T_Q :

- $T_Q = 6\ T_P$ *[1 mark]*

Determine the value of *W* by setting up simultaneous equations:

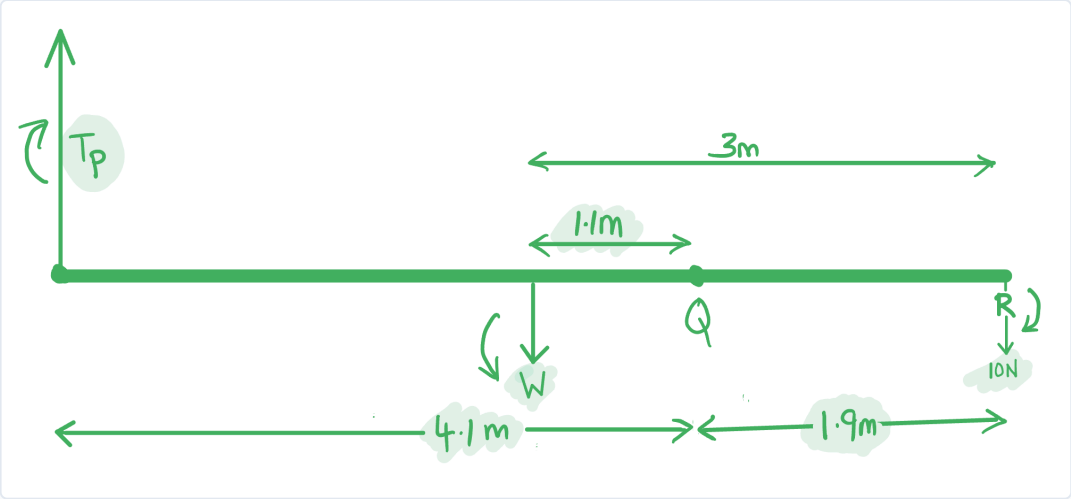
Apply the principle of moments around **P**:



- sum of the clockwise moments = sum of the anti-clockwise moments
- $3W + 10 \times 6 = 4.1 \times 6 \ T_P$
- $3W + 60 = 24.6 \ T_P$
- $T_P = \frac{3W + 60}{24.6}$ *(equation 1) [1 mark]*

Apply the principle of moments around **Q**:

Sketch a diagram showing measurements and forces:



- sum of the clockwise moments = sum of the anti-clockwise moments
- $10 \times 1.9 + 4.1 \ T_P = 1.1 \ W$
- $4.1 \ T_P = 1.1 \ W - 19$
- $T_P = \frac{1.1 \ W - 19}{4.1}$ *(equation 2) [1 mark]*

Combine *equations 1* and *2* to find *W*:

- $\frac{3 \ W + 60}{24.6} = T_P = \frac{1.1 \ W - 19}{4.1}$ *[1 mark]*

- $4.1 \times 3 \quad W + 60 \quad = 24.6 \times 1.1 \quad W \quad - \quad 19$
- $12.3 \quad W \quad + \quad 246 \quad = \quad 27.06 \quad W \quad - \quad 467.4$
- $27.06 - 12.3 \quad W \quad = \quad 246 + 467.4 \quad \rightarrow \quad 14.76 \quad W \quad = \quad 713.4$
- $W \quad = \quad 48 \quad \text{N} \quad 2 \quad \text{s.f.}$ *[1 mark]*

[Total: 5 marks]

*Without a clear sketch showing all the distances and forces, you have a **much** smaller chance of setting up the correct simultaneous equations, especially with the different distances. It also helps clearly see which forces are clockwise or anticlockwise depending on where you take moments from. Although the sketch doesn't earn marks directly, it is good practice for getting the calculation right.*

Part (b) [3 marks]

Q is now moved to a point further along the pole until the length **QR** is 0.8 m , as shown in Fig. 1.2.

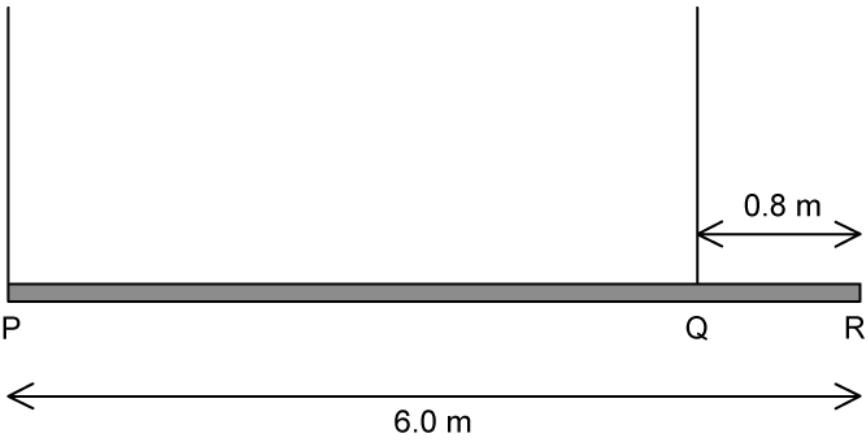


Fig. 1.2

The pole remains in equilibrium. The load of 10 N is now moved to a point *x* metres from **P**.

The beam remains in equilibrium in a horizontal position.

The rope at **Q** will break if its tension exceeds 32 N. The rope at **P** cannot break.

Find the range of possible positions on the beam where the load can be attached without the rope at **Q** breaking.

Mark Scheme (3 marks):

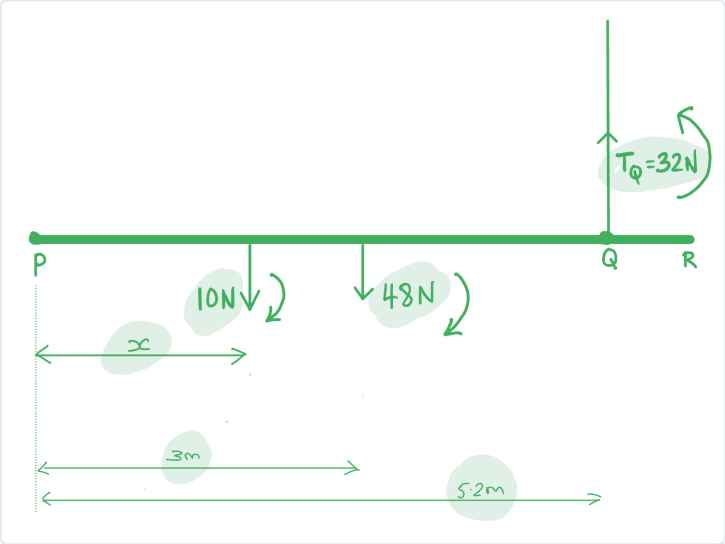
Find the range of possible positions on the beam where the load can be attached without the rope at Q.

List the known quantities:

- Weight of load, F = 10 N
- Distance **PR** = 6.0 m

- Distance **QR** = 0.8 m
- Weight of the pole, W = 48 N (from part a)
- Maximum tension in rope at **Q**, T_{Qmax} = 32 N

Make a sketch showing the forces and distances:



Apply principle of moments by taking moments about **P**:

- sum of clockwise moments = sum of anti-clockwise moments
- $10 \quad x \quad + \quad 3 \times 48 \quad = \quad 5.2 \quad \times T_{Qmax}$
- $10 \quad x \quad + 144 \quad = \quad 166.4$ *[1 mark]*
- $10 \quad x \quad = \quad 166.4 \quad - \quad 144 \quad = \quad 22.4 \quad \text{m}$
- $x \quad = \quad \frac{22.4}{10} = 2.24 \quad \text{m}$ *[1 mark]*

Deduce the range:

- $0 \quad \text{m} \quad \leq \quad x \quad \leq \quad 2.24 \quad \text{m}$ *[1 mark]*

[Total: 3 marks]

*Notice that the question asks for a **range**, meaning just an answer of 2.24 m will not get you full marks. x = 2.24 m is just the maximum distance before exceeding the tension limit of the rope at **Q**. The load could be placed at a smaller x (i.e. closer to **P**).*

If you used the unrounded answer for weight from part a), the calculated range would be $0 \quad \text{m} \quad \leq \quad x \quad \leq \quad 2.14 \quad \text{m}$ *which would be awarded a mark.*

Question 6 Medium [9 marks]

Part (a) [1 mark]

State what is meant by the centre of gravity of an object.

Mark Scheme (1 marks):

Centre of gravity is defined as...

- The point on an object at which a (resultant) force acts such that it will not create a moment / will not cause rotation**OR**The point at which all the (object's) mass / weight acts; *[1 mark]*

[Total: 1 mark]

*Careful with your wording, do **not** use definitions such as:*

- *Where mass is most concentrated*
- *It has the same mass on both sides*
- *All forces act through this point*

Part (b) [3 marks]

A uniform plank of wood of mass 40 kg and length 5.0 m is used by a boy to help him cross a ditch.

In the ditch is a rock, which is used to support the plank horizontally 1.1 m from one end, as shown in Fig. 1.1. The other end of the plank is supported by the bank.

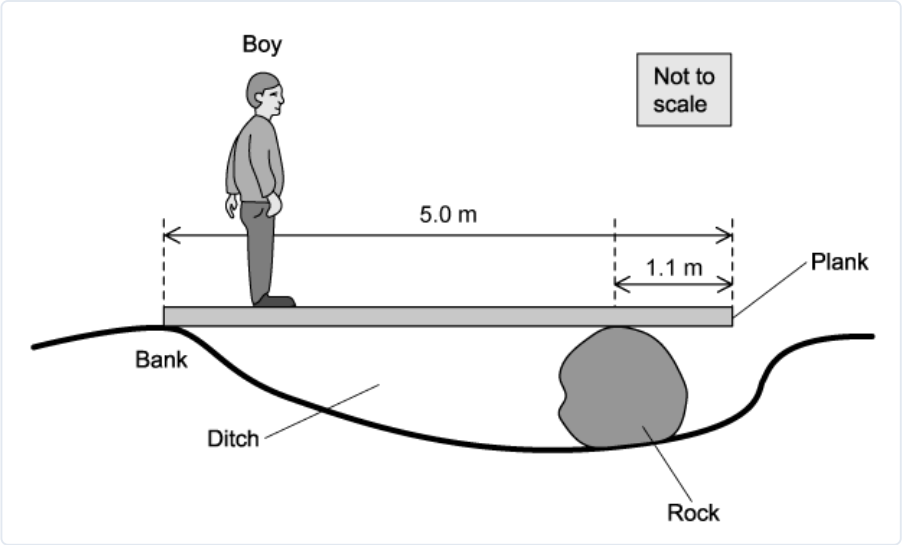


Fig. 1.1

Calculate the vertical supporting force from the rock when the plank is placed in position as shown in Fig 1.1.

Mark Scheme (3 marks):

Calculate the vertical supporting force from the rock:

- The weight of 40 × *g* occurs at the centre of the (uniform) plank so *d* = ½ × 5.0
- Moment of plank from the bank = *mg* × *d*
- Moment of plank from the bank = (40 × *g*) × (½ × 5.0)
- Moment of plank from the bank = 40 × 9.81 × 2.5 = 981 N m *[1 mark]*

This moment is balanced by the reaction force from the rock

- Moment of plank from the rock = *F* × (5.0 – 1.1)
- Moment of plank from the rock = *F* × 3.9 *[1 mark]*
- Vertical force from rock: $F = \frac{981}{3.9}$ = 251.5 = 250 N *[1 mark]*

[Total: 3 marks]

Part (c) [2 marks]

The boy begins to walk across the plank. The position of the boy is shown in Fig. 1.2.

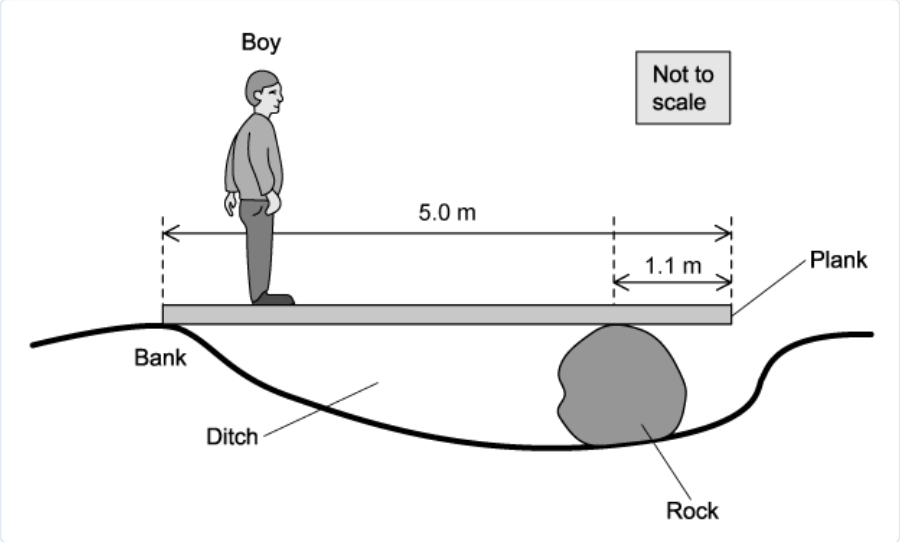
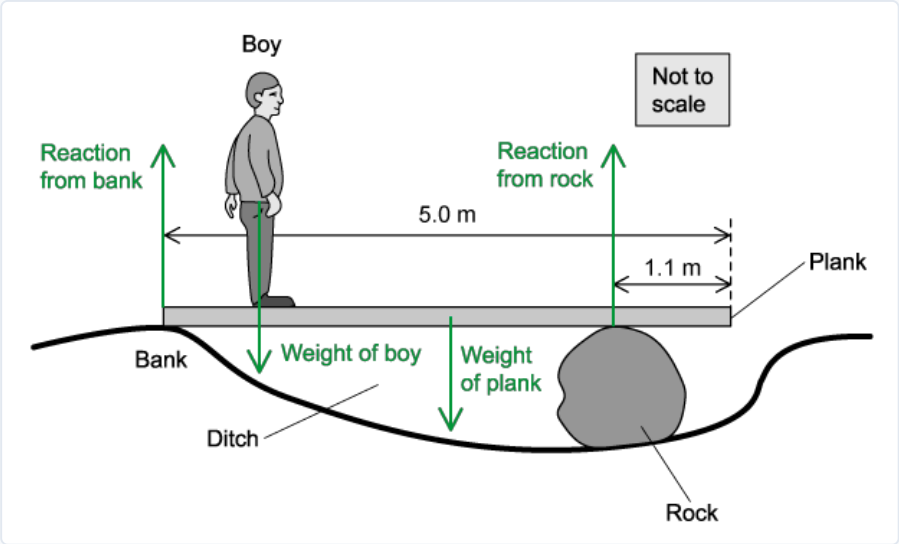


Fig. 1.2

On Fig. 1.2, draw arrows to show the forces acting on the plank when the boy has crossed some of the plank.

Mark Scheme (2 marks):

The forces acting on the plank when the boy has crossed some of the plank are:



- Correctly drawn arrows; *[1 mark]*
- Correctly labelled arrows; *[1 mark]*

[Total: 2 marks]

When labelling the weight of an object, make sure to draw from the centre of mass of the object.

Part (d) [3 marks]

The boy has a mass of 50 kg.

Determine whether the boy can walk to the far end of the plank without it tipping. Support your answer with a calculation.

Mark Scheme (3 marks):

Determine whether the boy can walk to the far end of the plank without it tipping:

- (At the point of tipping) there is no (reaction) force from the bank; *[1 mark]*

  This point must be in words, not implied from a calculation

- So, moments on the RHS of the rock must be lower than moments on the LHS

Supporting calculation 1:

- Taking moments about the rock to show RHS < LHS; *[1 mark]*
- 540 N m < 550 N m, therefore, plank will not tilt; *[1 mark]*

  Example of calculation:

  - LHS = (2.5 – 1.1) × 40 × g = 56 × 9.81 = 549.36 = 550 N m
  - RHS = 1.1 × 50 × g = 55 × 9.81 = 539.6 = 540 N m

OR Supporting calculation 2:

- Showing a moment calculation that gives the maximum boy's weight / mass that can be supported on the far edge of the plank (500 N / 51 kg) *[1 mark]*
- 50 kg < 51 kg, therefore, plank will not tilt; *[1 mark]*

  Example of calculation:

  - Maximum weight = $\frac{40 \times 9.81 \times 1.4}{1.1}$ = 499 = 500 N (or 51 kg)

OR Supporting calculation 3:

- Showing a moment calculation that gives the maximum distance the boy can be from the rock without tipping (1.12 m) *[1 mark]*
- 1.1 m < 1.12 m, therefore, plank will not tilt; *[1 mark]*

  Example of calculation:

  - Maximum distance = $\frac{40 \times 1.4}{50}$ = 1.12 m

[Total: 3 marks]

Question 7 Medium [9 marks]

Part (a) [2 marks]

It is said that Archimedes used huge levers to sink Roman ships invading the city of Syracuse.

A possible system is shown in Fig. 1.1 where a rope is hooked onto the front of the ship and the lever is pulled by several men.

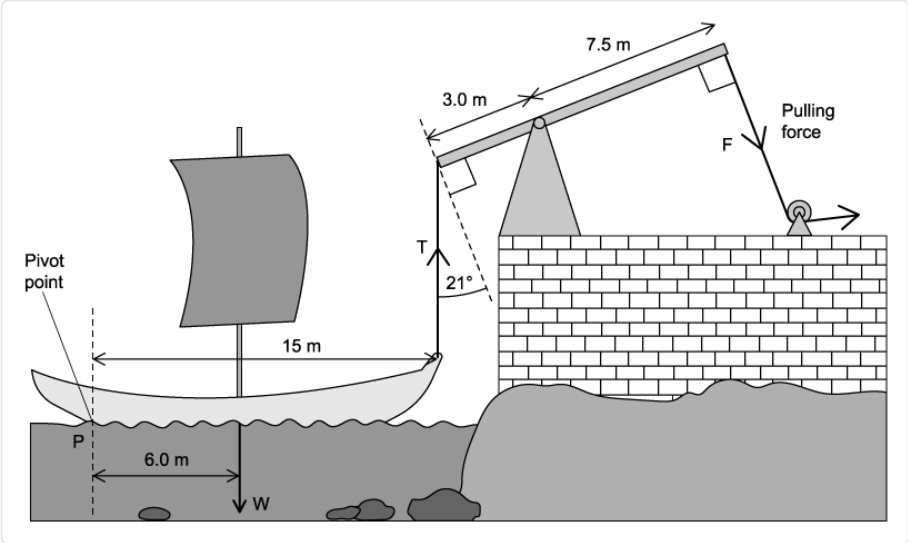


Fig. 1.1

Calculate the mass of the ship if its weight is 4.6×10^4 N. Give your answer to an appropriate number of significant figures.

Mark Scheme (2 marks):

Calculate the mass of the ship

- Weight: $W = mg \quad \Rightarrow \quad m = \frac{W}{g}$
- Mass: $m = \frac{4.6 \times 10^4}{9.81}$ = 4689 kg *[1 mark]*
- Mass: m = 4700 kg (2 s.f.) *[1 mark]*

[Total: 2 marks]

Part (b) [2 marks]

Calculate the moment of the ship's weight about point **P**. State an appropriate unit for your answer.

Mark Scheme (2 marks):

Calculate the moment of the ship's weight about point **P**:

- Moment about point **P** = Weight × Distance

- Moment = 46 000 × 6.0 = 2.8 × 10^5 *[1 mark]* N m *[1 mark]*

[Total: 2 marks]

When quoting a unit for moments, it is crucial to get the capitalisation correct i.e. do not write NM / nM / nm

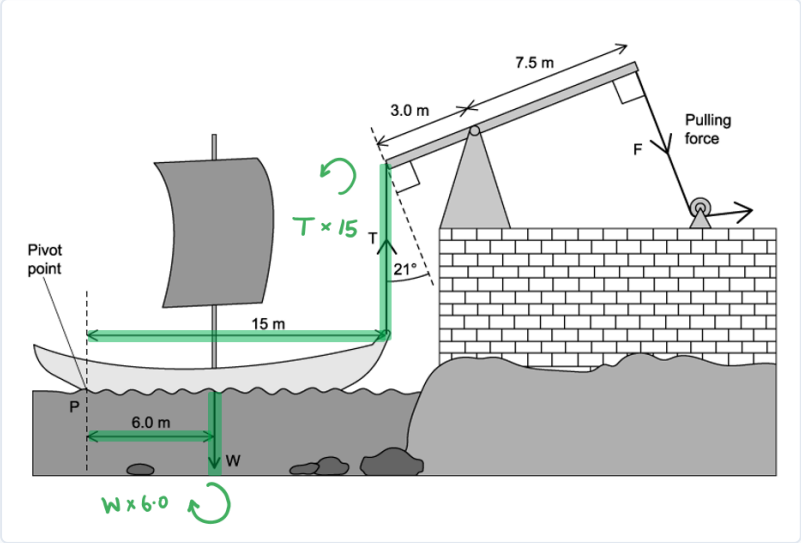
Part (c) [2 marks]

Calculate the minimum vertical force *T* required to start raising the front of the ship. Assume the ship pivots about point **P**.

Mark Scheme (2 marks):

Calculate the minimum vertical force *T*:

- Total clockwise moments (tension in rope) about P = *T* × 15
- Total anticlockwise moments (weight of the ship) about P = *W* × 6.0



Use the principle of moments to find tension *T*:

- Total clockwise moments = Total anticlockwise moments
- *T* × 15 = (4.6 × 10^4) × 6.0 *[1 mark]*
- $T = \frac{4.6 \times 10^4 \times 6.0}{15}$ = 1.8 × 10^4 N *[1 mark]*

Allow T = 1.9 × 10^4 N if 2.8 × 10^5 used

[Total: 2 marks]

Part (d) [3 marks]

Calculate the minimum force *F* that must be exerted to start to raise the front of the ship.

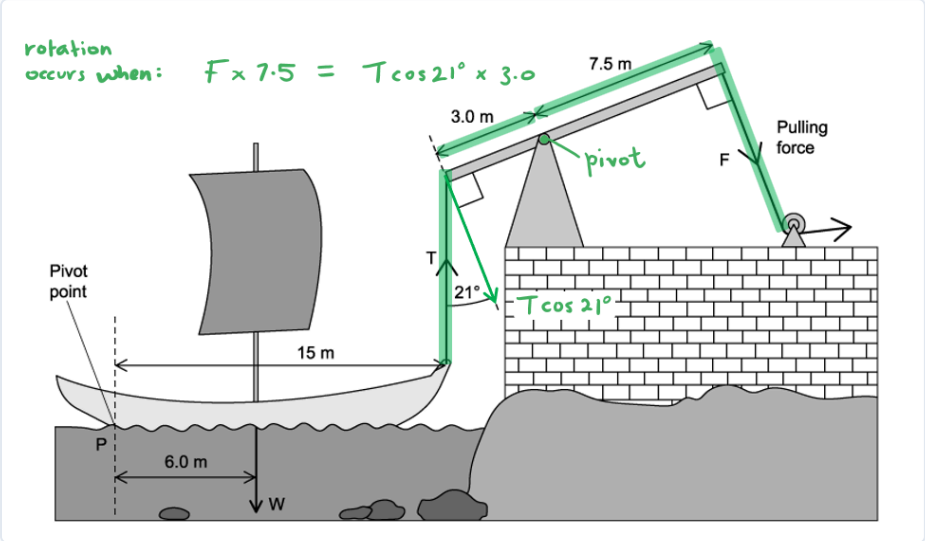
Mark Scheme (3 marks):

Calculate the minimum force *F*:

- Perpendicular component of *T* = *T* cos (21°) *[1 mark]*

The front of the ship will raise when...

- Total clockwise moments (pulling force *F*) > Total anticlockwise moments (perpendicular component of *T*)
- *F* × 7.5 > *T* cos (21°) × 3.0



The minimum force required to raise the ship is...

- Total clockwise moments = Total anticlockwise moments
- *F* × 7.5 = *T* cos (21°) × 3.0 *[1 mark]*
- $F = \frac{1.8 \times 10^4 \cos 21° \times 3.0}{7.5}$ = 6722 = 6700 N (2 s.f.) *[1 mark]*

Allow F = 7100 N if 1.9 × 10^4 used

[Total: 3 marks]

Question 8 Medium [10 marks]

Part (a) [1 mark]

State what is meant by the *centre of gravity* of an object.

Mark Scheme (1 marks):

The centre of gravity of an object is:

- The point where the (entire) weight (of an object) is considered to act *[1 mark]*

Part (b) [2 marks]

State the **two** conditions required for an object to be in equilibrium.

Mark Scheme (2 marks):

The two conditions required for an object to be in equilibrium are:

- Resultant force is zero *[1 mark]*
- Resultant moment / torque (about any point) is zero *[1 mark]*

Part (c) [5 marks]

A uniform beam AB of length 2.40 m and weight 50 N is hinged at end A. The beam is held in a horizontal position by a vertical wire attached to end B, as shown in Fig. 2.1.

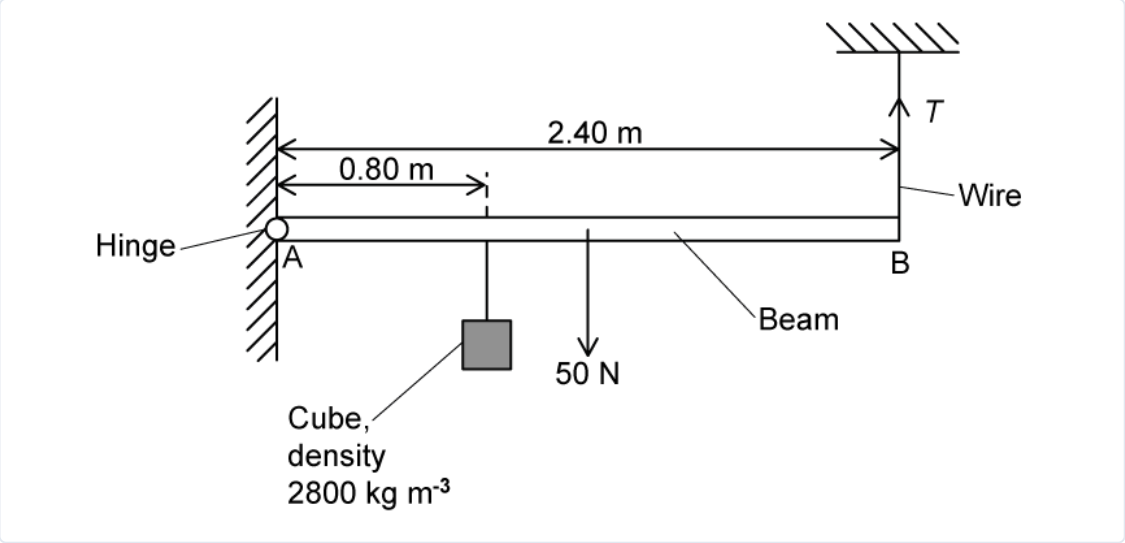


Fig. 2.1 (not to scale)

A cube of side length 0.075 m is suspended from the beam at a distance of 0.80 m from A. The cube is made from a material of density 2800 kg m^{-3}.

The beam is in equilibrium. Assume that the hinge is frictionless.

(i) Show that the weight of the cube is about 12 N.

[2]

(ii) By taking moments about A, calculate the tension T in the wire.

[2]

Mark Scheme (5 marks):

(i) To show that the weight of the cube is about 12 N:

- List the known quantities
  - Side length of the cube, l = 0.075 m
  - Density of the cube, ρ = 2800 kg m^{-3}
  - Acceleration of free fall, g = 9.81 m s^{-2}
- Calculate the volume of the cube:

$$V = l^3$$

$$V = 0.075^3 \text{ } \textit{[1 mark]}$$

- Use the density equation to calculate the weight of the cube:

$$\rho = \frac{m}{V}$$

$$W = mg = \rho V g$$

$$W = 2800 \times 0.075^3 \times 9.81$$

$$W = 11.58...$$

$$W = 12 \text{ N (2 s.f.) } \textit{[1 mark]}$$

(ii) To calculate the tension T in the wire:

- Identify the moments acting about A

Anticlockwise moment = $T \times 2.40$

Clockwise moments = $50 \times 1.20 + 11.58 \times 0.80$ *[1 mark]*

- Apply the principle of moments and calculate T:

Since the beam is in equilibrium, anticlockwise moments = clockwise moments

$$T \times 2.40 = 50 \times 1.20 + 11.58 \times 0.80$$

$$T = \frac{50 \times 1.20 + 11.58 \times 0.80}{2.40}$$

$$T = 28.86...$$

$$T = 29 \text{ N } \textit{[1 mark]}$$

1 mark for $T \times 2.40$ **OR** 50×1.20 **OR** 11.58×0.80

1 mark for the correct application of the principle of moments to calculate T

Part (d) [2 marks]

The cube in **(c)** is now completely submerged in a beaker of water, as shown in Fig. 2.2.

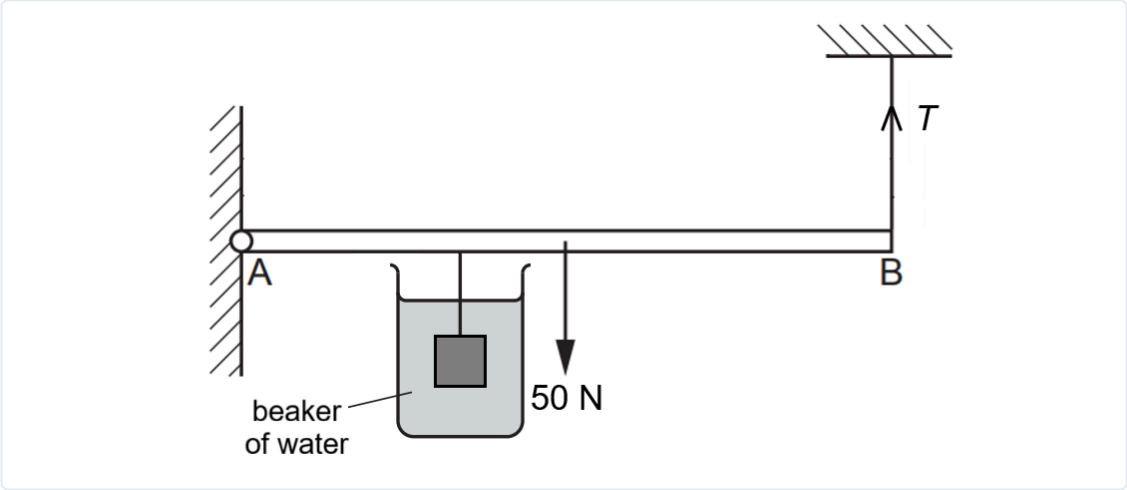


Fig. 2.2

The magnitude of T is unchanged.

State and explain the effect of this change on the magnitude and direction of the resultant moment of the forces on the beam about A.

Mark Scheme (2 marks):

State the effect of this change on the magnitude and direction of the resultant moment of the forces on the beam about A:

- Resultant/net moment is anticlockwise (no longer zero) *[1 mark]*

Explanation:

- The clockwise moment decreases as the apparent weight of the cube decreases *[1 mark]*

It would be incorrect to say that the anticlockwise moment increases due to upthrust acting on the cube. This is because the upthrust does not act directly on the beam; therefore, it does not introduce a new clockwise moment about A on the beam; it simply reduces the tension in the cube's support. Hence, it is the clockwise moment on the beam that decreases.

Question 9 Hard [12 marks]

Part (a) [4 marks]

In a double-decker bus there are two passenger decks, stacked one above the other.

The dimensions of an older, 'Routemaster' bus are:

Height = 4.38 m

Width = 2.44 m

Mass = 7.47 tons

To ensure safety, buses are tested for stability using a tilting platform as shown in Fig. 1.1.

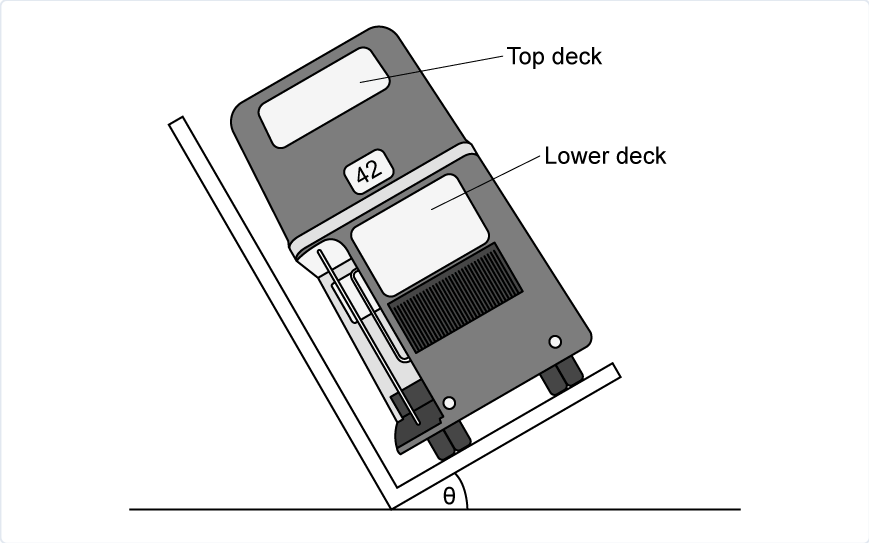


Fig. 1.1

Weights are attached to the underside of the bus, then the angle the bus makes with the horizontal is gradually increased until the bus begins to topple to the left.

(i) State the reason that the bus would topple to the left.

[1]

(ii) Explain the effect of adding weights to the underside of the bus.

[3]

Mark Scheme (4 marks):

(i) State the reason that the bus would topple to the left:

The bus begins to topple because:

- The line of action of the centre of gravity falls outside the base of the bus **OR** the anticlockwise moment (of the bus) is greater than the clockwise moment; *[1 mark]*

(ii) Explain what effect adding weights to the underside of the bus would be:

- Lower the centre of gravity; *[1 mark]*
- It could tilt more before toppling (to the left); *[1 mark]*
- Increases stability; *[1 mark]*

[Total: 4 marks]

Part (b) [2 marks]

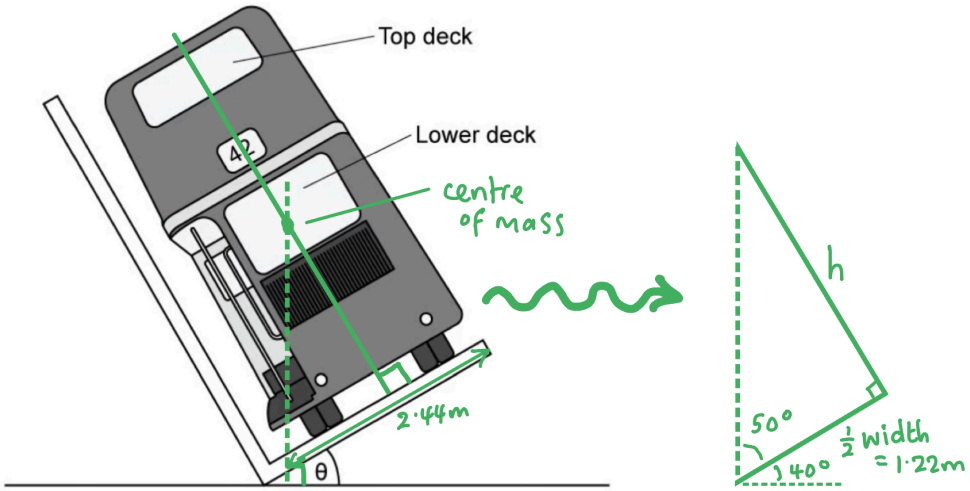
The bus is first tested completely empty, and is found to tilt to an angle of 40° from the horizontal before it starts to tip.

Determine the height above the ground of the centre of mass of the bus when it is empty and normally parked.

Mark Scheme (2 marks):

Determine the height above the ground of the centre of mass of the bus when it is empty and normally parked:

Redraw the diagram as a triangle:



Either on the diagram or seen in later calculation

- Angle to be used in calculation = 50° **OR** short side of triangle = 1.22 m *[1 mark]*

Calculate the height of the centre of mass:

- $h = 1.22 \times \tan 50 = 1.5$ m *[1 mark]*

[Total: 2 marks]

Since A-level questions are more complex there are often 'intermediary' or 'working' marks for numbers seen within a calculation. In this case you are being awarded one mark for correctly deriving at least one of the values you will need to calculate.

Your working is being rewarded so it's well worth showing it.

The bus is not of uniform density, so it's centre of mass will not be in the centre of the object.

Part (c) [3 marks]

To be considered safe to use on the road the bus must be safe when carrying passengers.

To test this the top deck is weighted with iron blocks or sandbags to the equivalent of 60 passengers, while the lower deck is left empty.

The average passenger is assumed to have a mass of 65 kg and their centres of mass can be considered to act at a height of 3.28 m above the road.

1 ton = 1000 kg

For this safety test

(i) Calculate the new height of the centre of mass of the bus above the road.

[2]

(ii) Suggest why the test is done with the top deck weighted to capacity and the bottom deck left empty.

[1]

Mark Scheme (3 marks):

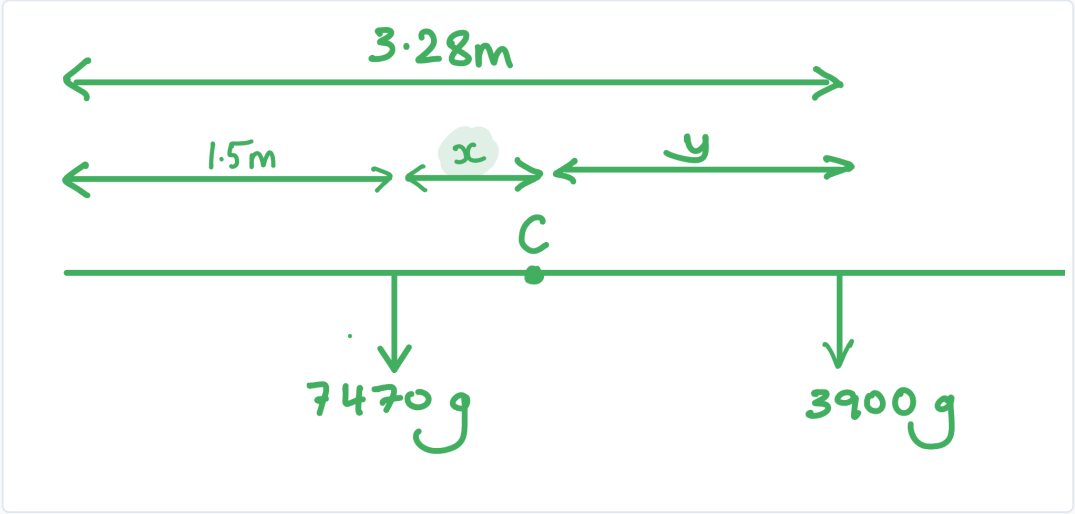
(i) Calculate the new height of the centre of mass of the bus:

Determine the total mass of the 'passengers' and the empty bus:

- Mass of 'passengers', $m_p = 65 \text{ kg} \times 60 \text{ passengers} = 3900 \text{ kg}$
- Mass of empty bus, $m_b = 7.47 \text{ tons} \times 1000 \text{ kg} = 7470 \text{ kg}$

Model the bus as a horizontal rod and take moments:

Method 1: take moments around the new centre of mass and calculate:



Set up simultaneous equations for *x* and *y*:

- In equilibrium, $7470g \times x = 3900g \times y \rightarrow 7470x = 3900y$ (*equation 1*)
- Taking distance measurements, $1.5 + x + y = 3.28 \rightarrow y = 1.78 - x$ (*equation 2*)

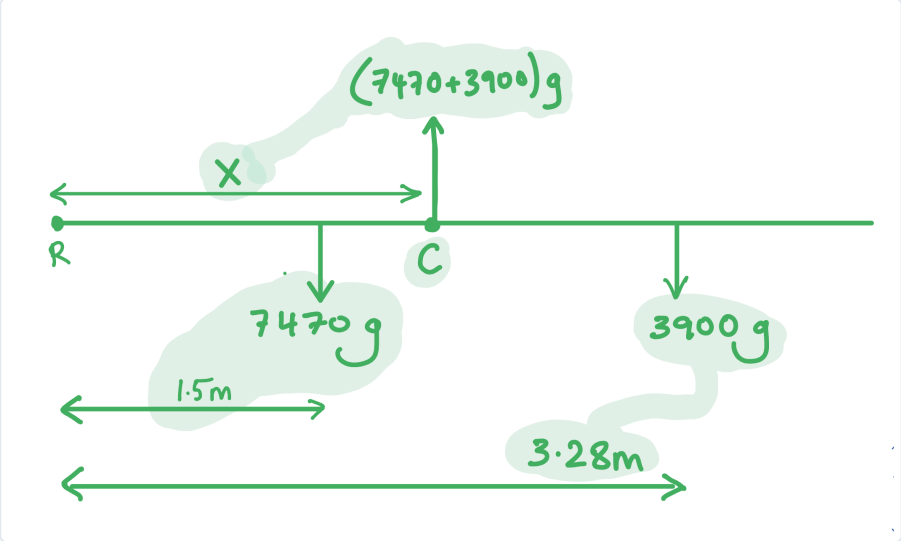
Substitute *equation 2* into *equation 1* and calculate:

- $7470\ x = 3900y$
- $7470\ x = 3900 \times 1.78 - x$
- $7470\ x + 3900x = 6942$
- $x = 0.61$ m *[1 mark]*

Add distances together to find the height of C above the road:

- new height, $h = 1.5 + 0.61 = 2.11$ m *[1 mark]*

Method 2: take moments about the road, assuming a combined mass, and calculate:



Apply the principle of moments around **R**::

- In equilibrium, $7470g \times 1.5 + 3900g \times 3.28 = 7470g + 3900g \times X$ *[1 mark]*

Calculate:

- $11\ 205g + 12\ 792g = 11\ 370g \times X$
- $X = 2.11$ m *[1 mark]*

(ii) Suggest why the test is done with the top deck weighted to capacity and the bottom deck left empty:

- This moves the centre of mass to the highest position / makes the bus the least stable it could be; *[1 mark]*

[Total: 3 marks]

Part (d) [3 marks]

To pass the safety test the bus must be able to tilt to 28° when loaded as in part (c).

(i) Deduce whether the bus has passed the safety test.

[2]

(ii) The safety testers imagine that the bus had 60 passengers upstairs, no passengers downstairs and that **all** the passengers on the top deck stood up simultaneously. This would shift their centres of mass by a mean distance of 40 cm higher.

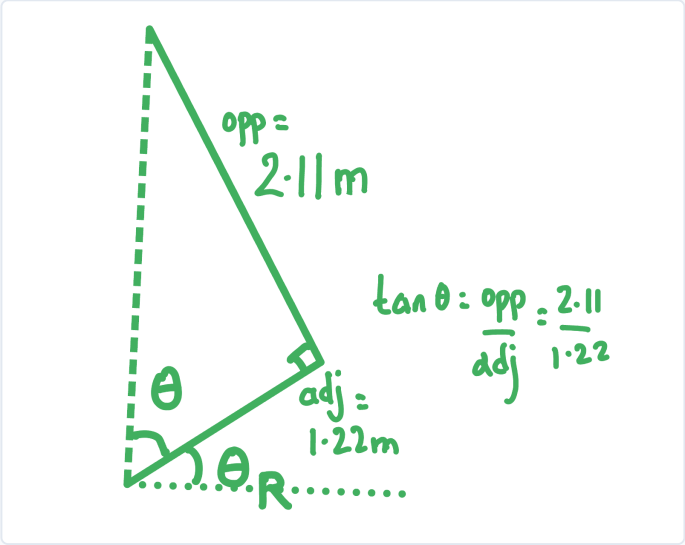
Explain whether this will make the bus fail its safety test.

[1]

Mark Scheme (3 marks):

(i) Deduce whether the bus has passed the safety test:

Set up a triangle to calculate the angle using trigonometry:



- Angle inside triangle, $\theta = \tan^{-1}\frac{2.11}{1.22} = 60°$
- Angle between bus and road, $\theta_R = 90 - 60 = 30°$ *[1 mark]*

Draw a conclusion:

- The bus tilted to more than 28° without tipping so yes it does pass the safety test; *[1 mark]*

(ii) Explain whether all the passengers upstairs standing will make the bus fail its safety test.:

Calculate new angle between bus and road by moving the centre of mass 40 cm higher:

- $\theta = \tan^{-1}\frac{2.11 + 0.4}{1.22} = 64°$
- New angle between bus and road, $\theta_R = 90 - 64 = 26°$

Draw a conclusion:

- The bus tilts to less than 28° without tipping so yes it will fail the safety test; *[1 mark]*

[Total: 3 marks]

For the final part of the question you could have recalculated the new centre of mass due to the passengers standing up. However, for only one mark, there is probably a quicker way.

By simply moving the centre of mass up by the full 40 cm (which is the maximum it could possibly shift) we still find that the bus fails the safety test, so the question has been answered without the need for further calculation.

Question 10 Hard [15 marks]

Part (a) [4 marks]

A bench consists of a plank which is resting in a horizontal position on two supports B and C. The plank is modelled as a uniform rod AD of length 2.7 m and mass 20 kg. The supports at B and C are 0.5 m from each end of the plank, as shown in Fig. 1.1.

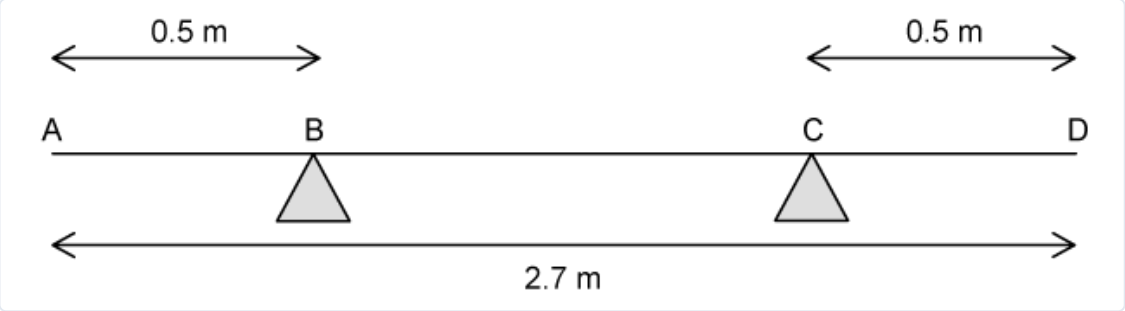


Fig. 1.1

Two animals sit on the plank. The rabbit has a mass of 2.5 kg and sits in the middle of the plank and the quail has a mass of 4.3 kg and sits at the end A. The plank remains horizontal and in equilibrium. The animals can be modelled as point masses.

Calculate:

(i) The magnitude of the normal reaction force at B

[2]

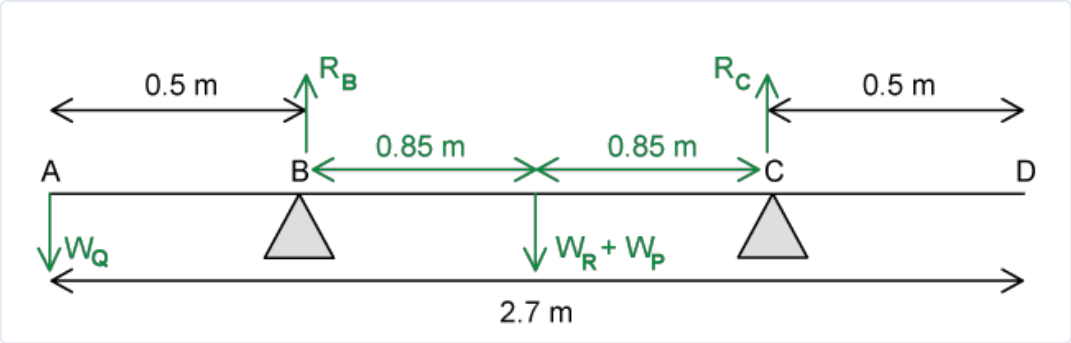
(ii) The magnitude of the normal reaction force at C

[2]

Mark Scheme (4 marks):

List the known quantities:

- Mass of rabbit, m_R = 2.5 kg
- Mass of quail, m_Q = 4.3 kg
- Mass of the plank, m_P = 20 kg
- Acceleration due to gravity, g = 9.81 m s^{-2}



(i) The magnitude of the normal reaction force at B

Take moments about point C:

- Principle of moments: Sum of anti–clockwise moments = sum of clockwise moments about a pivot
- Anti–clockwise moments about C = ($W_Q \times d_Q$ + ($W_R + W_P$) × d_P) where d_C and d_P are perpendicular distances from the pivot to the weight of the quail and the plank, respectively
- Clockwise moment about C = $R_B \times d_B$

Calculate the normal reaction force at B, R_B :

- (4.3g × (2.7 – 0.5)) + ((2.5g + 20g) × (0.85) = R_B × (0.85 + 0.85) *[1 mark]*
- 9.46g + 19.125g = R_B × 1.7
- $R_B = \frac{9.46g + 19.125g}{1.7}$ = 16.81g = 16.81 × 9.81 = 164.9 = 165 N *[1 mark]*

(ii) The magnitude of the normal reaction force at C

Equate vertical forces:

- Since the plank is in equilibrium, all the forces upwards = all forces downwards
- Forces upwards = $R_B + R_c$
- Forces downwards = $W_Q + W_R + W_P$

Calculate normal reaction force at C, R_c :

- 16.81g + R_c = 4.3g + 2.5g + 20g = 26.8g *[1 mark]*
- R_c = 26.8g – 16.81g = 9.99g = 9.99 × 9.81 = 98.00 = 98 N (2 s. f.) *[1 mark]*

[Total: 4 marks]

This question gives the masses of the animals instead of their weight. Whether you multiply by g to get the weight during the calculation or at the end is up to you. In some instances where there are a lot of calculations, it is easier to only multiply by g (to calculate their weight) at the very end to avoid numerical errors.

Students often forget about the fact that the vertically upwards forces must also equal the vertically downwards forces for a system to be in equilibrium too. Therefore, you do not need to take moments again for part (ii).

Part (b) **[4 marks]**

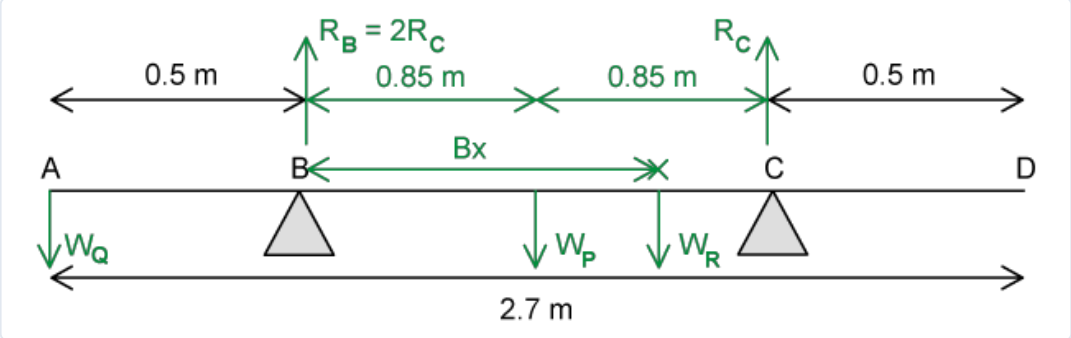
Whilst the quail stays at point A the rabbit now moves along the plank to point X.

Given that the plank remains horizontal and in equilibrium, and that the magnitude of the normal reaction force at B is now twice the magnitude of the normal reaction force at C, calculate the distance BX.

Mark Scheme (4 marks):

List the known quantities:

- Mass of rabbit, m_R = 2.5 kg
- Mass of quail, m_Q = 4.3 kg
- Mass of the plank, m_P = 20 kg
- Acceleration due to gravity, g = 9.81 m s^{-2}



Equate vertical forces:

- Since the plank is in equilibrium, all the forces upwards = all forces downwards
- Forces upwards = $R_B + R_c = 2R_c + R_c = 3R_c$
- Forces downwards = $W_Q + W_R + W_P$

Calculate normal reaction force at C, R_c:

- $3R_c$ = 4.3g + 2.5g + 20g = 26.8g *[1 mark]*
- $R_c = \frac{26.8g}{3}$ = 8.933g (= 87.636 N) *[1 mark]*

Take moments about B:

- Anti–clockwise moments about B = ($W_Q \times d_Q$) + ($R_c \times d_c$)
- Clockwise moments about B = ($W_P \times d_P$) + ($W_R \times d_R$)

Calculate distance BX, d_R:

- (4.3g × 0.5) + (8.933g × (0.85 + 0.85) = (20g × 0.85) + (2.5g × d_R) *[1 mark]*
- 2.15g + 15.1861g = 17g + (2.5g × d_R)
- d_R = BX = $\frac{2.15g + 15.1861g - 17g}{2.5g}$ = 0.13 m *[1 mark]*

[Total: 4 marks]

Labelling all the forces and distances on the diagram in your exam is essential for moments questions. This helps calculate any missing distances.

Whenever any point moves in a moments question, the reaction forces upwards will also change. Therefore, R_c and R_B are not the same as in part (a) for this part (b).

The point where the pivot is taken from doesn't actually matter (because the Principle of Moments will always be true, about any pivot), so you will still get the same final answer. However, since we do not know in this question (it is different to part (a)), then if we take moments about B then we do not need to worry about this force. The key is to take moments about a point where you don't know the value of the force: this way, the unknown force will not appear in your moments equations (since d will be 0).

Part (c) **[4 marks]**

A pole PR has a length 4 m and a weight *W* newton. The pole is held in a horizontal position in equilibrium by two vertical ropes attached to the pole at the points P and R where PR = 4 m, as shown in Fig. 1.2.

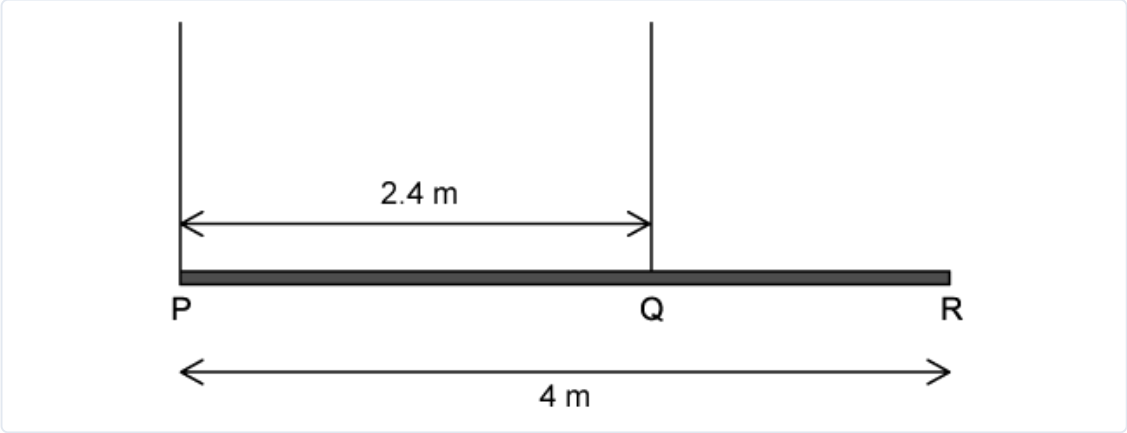


Fig. 1.2

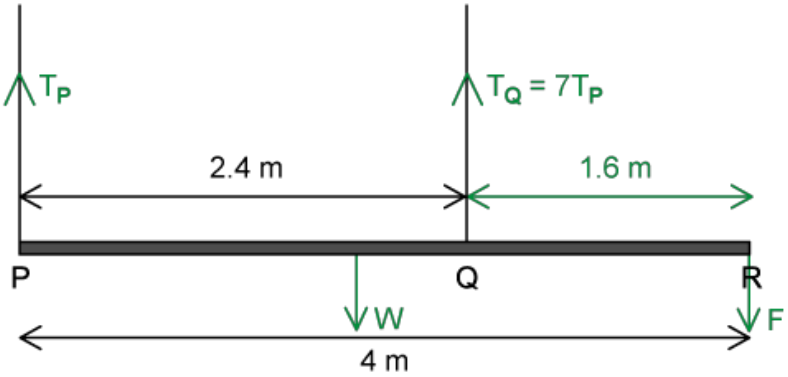
A load of weight 17 N is attached to the rod at R. The pole is uniform and the ropes are light inextensible strings.

Given that the tension in the rope attached to the pole at Q is seven times the tension in the rope attached to the pole at P, calculate the value of *W*.

Mark Scheme (4 marks):

List the known quantities:

- Weight of load, *F* = 17 N
- Distance PR = 4 m
- Distance QR = 4 − 2.4 = 1.6 m
- $T_Q = 7T_P$



Equate vertical forces:

- Since the plank is in equilibrium, all the forces upwards = all forces downwards
- Forces upwards: $T_P + T_Q = T_P + 7T_P = 8T_P$
- Forces downwards: $W + F$
- $8T_P = W + F = W + 17$ *[1 mark]*

Determine the equation for the tension in the rope at P, T_P:

- Taking moments about point Q
- Anti–clockwise moment about Q = $(W \times d_W)$ where d_W is the distance of the weight W from Q
- Clockwise moments about Q = $(T_P \times d_P) + (F \times d_F)$ where d_P and d_F are the distance of P and the force from the load F from Q respectively
- $(W \times (2.4 - 2)) = (T_P \times 2.4) + (17 \times 1.6)$
- $0.4W = 2.4T_P + 27.2$ *[1 mark]*

Calculate the weight of the pole, W:

- $8T_P = W + 17$ and $0.4W = 2.4T_P + 27.2$
- $T_P = \frac{W}{8} + 2.125$
- $0.4W = 2.4(\frac{W}{8} + 2.125) + 27.2 = 0.3W + 5.1 + 27.2$ *[1 mark]*
- $0.1W = 32.3$
- $W = 323$ N *[1 mark]*

[Total: 4 marks]

Look out for simultaneous equations! This is where you can set up the same number of unique equations as there are unknown variables. In this question, there are two equations with two unknowns (W and T_P).

If you'd like to refresh your maths skills on this topic, then check out the Save My Exams GCSE Maths revision notes on 'Linear Simultaneous Equations'.

Part (d) **[3 marks]**

Q is now moved to a point further along the pole until the length QR is 0.8 m, as shown in Fig. 1.3. The pole still remains in equilibrium. The load of 17 N is now moved to a point *x* metres from P. The beam remains in equilibrium in a horizontal position.

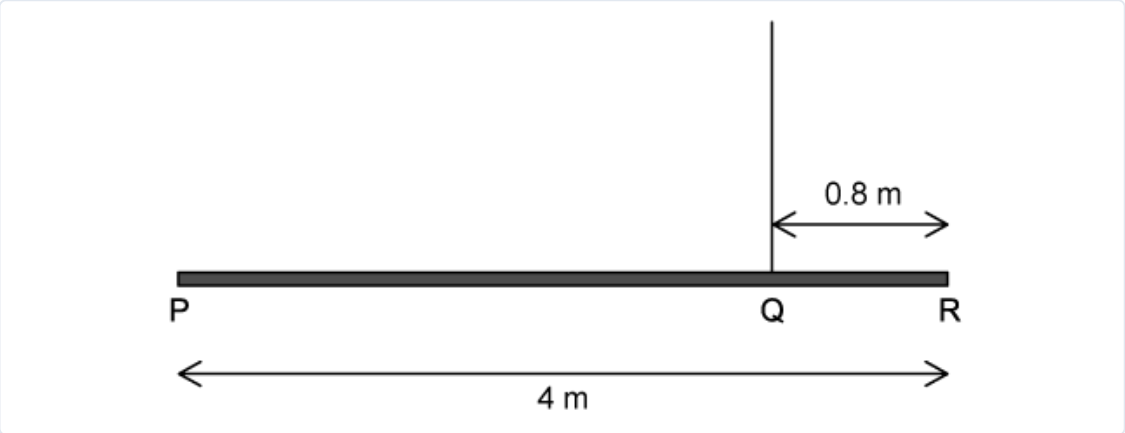


Fig. 1.3

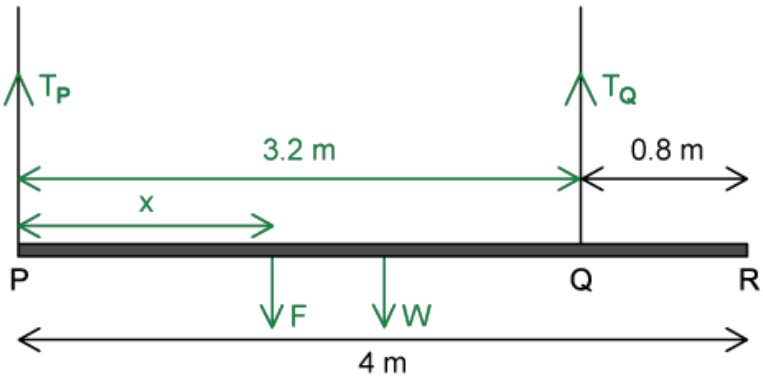
The rope at Q will break if its tension exceeds 212 N. The rope at P cannot break.

Find the range of possible positions on the beam where the load can be attached without the rope at Q breaking.

Mark Scheme (3 marks):

List the known quantities:

- Weight of load, *F* = 17 N
- Distance PR = 4 m
- Distance QR = 0.8 m
- Distance PQ = 3.2 m
- Weight of the pole, *W* = 323 N (from part c)
- Max T_Q = 212 N



Determine equation for tension at Q, T_P :

- Take moments about P
- Anti–clockwise moment about P = $T_Q \times d_Q$ where d_Q is the distance of Q from P
- Clockwise moments about P = ($F \times d_F$) + ($W \times d_W$) where d_F and d_W are the distance of the force *F* and the weight *W* from P respectively
- T_Q × 3.2 = 17*x* + (323 × 2) *[1 mark]*
- 3.2 T_Q = 17*x* + 646
- T_Q = 5.31*x* + 202 *[1 mark]*

Calculate the range of values for *x*:

- T_Q < 212 therefore, 5.31*x* + 202 < 212
- *x* < 1.88
- 0 < *x* < 1.88 m *[1 mark]*

[Total: 3 marks]

When you are asked for a 'range' of values, there is likely to be inequality. These are equations with the 'less than' (<) or 'more than' (>) symbols, and they can be rearranged exactly like regular equations. Just remember if you multiply or divide both sides of an inequality by a negative number, the direction of the inequality sign switches!

The details of the rearranging for the inequality are shown below:

$$5.31x + 202 < 212$$

Subtract 202 from both sides:

$$5.31x < 10$$

Divide both sides by 5.31:

$$x < 1.88$$